

KUMBHSETU 2026

Maha Kumbh Mela Nashik–Trimbakeshwar — 4-App Connected Digital Ecosystem

Document: System Architecture & User Guide	Target Event: Maha Kumbh Mela 2026
Backend Cloud: Supabase (PostgreSQL + Realtime)	Ecosystem Status: 4 Apps Synchronized & Verified

1. Executive Summary & Architecture

KumbhSetu 2026 is a multi-role digital ecosystem connecting millions of pilgrims, district administration authorities, local food & craft merchants, and ground safety volunteers during the Maha Kumbh Mela. All applications communicate through a unified Supabase cloud database with sub-second real-time sync, resilient memory-fallback offline storage, and automated ratings computation.

Application	Primary Role	Key Modules & Capabilities
KumbhSetu-Expo (Pilgrim Mobile App)	Yatris & Devotees	Live RTO transit fares, verified local bazaar shops & ratings, grievance reporting, rumor buster, snan muhurat & map.
KumbhSetu-Admins (Admin Portal)	RTO, Municipal Corp, Police Control	Set transit route price caps, essential commodity ceilings, audit & approve/reject/delete bazaar stalls, dispatch fact-checks.
KumbhSetu-LocalBazaar (Merchant App)	Local Vendors & Eateries	Register stall, upload FSSAI license & photos, 1-tap Open/Closed toggle, manage food catalog, customer inquiry desk.
KumbhVeer (Volunteer App)	Ground Patrol & Fact-Checkers	Sector-targeted incident alerts, on-the-spot physical inspection, Swipe-to-Resolve, ground rumor investigation (EN/MR).

2. Real-Time Cross-App Workflows

A. Transit Tariffs & Price Ceiling Enforcement

- **Admin Creation:** District Administration & RTO configure route tariffs in `tariff_routes` (Shared Auto, Bus, Private Auto, Taxi) and ceiling prices in `commodity_prices`.
- **Pilgrim Stream:** KumbhSetu-Expo streams the official route cards dynamically with real-time search, travel duration, distance, and 4-mode pricing.
- **Resilient Fallbacks:** API service includes automatic multi-tier query retries to ensure fares always display without failure.

B. Local Bazaar, Admin Auditing & Shop Ratings

- **Stall Registration:** Merchants register their eatery or stall with photos, address, and FSSAI badge in KumbhSetu-LocalBazaar. Initial state is set to pending verification.
- **Admin Authority:** In KumbhSetu-Admins, the administration audits the stall with full authority to **Approve & Verify** (publishes to Yatri app), **Reject/Revoke**, or **Delete** permanently.
- **Yatri Marketplace Visibility:** Only Admin-verified stalls (`is_verified = true`) appear in the Pilgrim marketplace.
- **Live Open/Closed Status:** Vendors toggle stall open/closed status in 1-tap; green 'OPEN' or gray 'CLOSED' badge updates immediately.
- **Rating & Dynamic Ranking:** Pilgrims submit 1–5★ reviews -> triggers database recalculation of average rating -> shops automatically rank with top-rated rising to the top.

C. Grievance Reporting & Volunteer Resolution

- **Citizen Report:** Pilgrims file overcharging or crowd safety complaints in KumbhSetu-Expo.
- **Squad Dispatch:** Realtime alerts notify both KumbhSetu-Admins and KumbhVeer volunteers in that duty sector.
- **Ground Resolution:** Volunteers visit the spot, add resolution notes, and **Swipe-to-Resolve** in KumbhVeer.

D. Rumor Buster & Fact-Checking

- **Claim Submission:** Pilgrims submit unverified social media rumors in the app.
- **Ground Investigation:** KumbhVeer volunteers physically verify conditions on the ground.
- **Official Broadcast:** Admins publish verified status (TRUE / FALSE) with official explanations live across all apps.

3. Supabase Cloud Database Schema

The backend comprises 9 relational tables with row-level security, triggers, and realtime publication:

Table Name	Primary Columns & Schema	Sync Scope
public.tariff_routes	id, route_name, standard_rate, shared_auto_rate, bus_rate, night_rate, distance_km, approx_minutes, traffic_note	Admin -> Pilgrims
public.commodity_prices	id, item_name, category, max_retail_price, unit, notes	Admin -> Pilgrims
public.merchants	id, business_name, owner_name, phone, category, scale_type, fssai_number, facade_image_url, is_verified, is_open_now, rating, review_count	Merchants <-> Admins <-> Pilgrims
public.catalog_items	id, merchant_id, name, category, price, description, image_url, is_available	Merchants -> Pilgrims
public.shop_reviews	id, merchant_id, user_name, rating, comment, created_at	Pilgrims -> Merchants/Admins
public.incidents_and_grievances	id, title, description, category, sector, status, priority, reporter_name, assigned_volunteer_name, resolution_notes, resolved_at	Pilgrims -> Admins & Volunteers
public.fact_checks_and_rumors	id, claim_title, verdict, official_explanation, submitted_by, verified_by_volunteer, verified_by_admin	All 4 Applications
public.volunteer_profiles	id, full_name, phone, id_card_number, assigned_sector, is_active, tasks_resolved_count	Volunteers <-> Admins
public.pilgrim_inquiries	id, merchant_id, item_id, pilgrim_name, pilgrim_phone, message, status	Pilgrims -> Merchants

4. Local Execution & Credentials

All 4 applications run on Expo SDK 57 with web, Android, and iOS support.

App Directory	Start Command	Default Login / Passcode
KumbhSetu-Expo	cd KumbhSetu-Expo && npx expo start	Open registration (Name, Phone, DOB) or Direct Sign-In
KumbhSetu-Admins	cd KumbhSetu-Admins && npx expo start	Username: Gaurang Password: pass123 (or PIN: 1008)
KumbhSetu-LocalBazaar	cd KumbhSetu-LocalBazaar && npx expo start	Register new stall or enter 10-digit mobile number
KumbhVeer	cd KumbhVeer && npx expo start	Register volunteer or badge ID (e.g. KV-RAMKUND-01)

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